

Bishal Dhungana

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SUMMARY

Data Scientist with hands-on experience building and shipping computer vision and generative AI systems, plus the data pipelines and cloud infrastructure that support them. I independently built and deployed a switch-state classification model to production on AWS SageMaker/ECR, benchmarked detection/segmentation architectures (YOLO, RT-DETR, Detectron2) for transfer learning, and defined ML scope for a multi-vendor CV engagement. Also hands-on with agentic AI systems (RAG, MCP, LLM tool-use) and LangChain.

SKILLS

Machine Learning & Computer Vision: PyTorch, TensorFlow/Keras, Ultralytics, Roboflow, YOLO, RT-DETR, Detectron2, Faster/Mask R-CNN, EfficientDet, DETR, MMDetection, ResNet, XGBoost, OpenCV, ONNX

Generative AI & Agentic Systems: LangChain, LangGraph, RAG, Model Context Protocol, FastMCP, Vector Databases, Ollama, Chroma, Hugging Face Transformers

Data Engineering: Python, SQL, ETL/ELT Pipelines, PySpark, DuckDB, Databricks, Data Lakes, Parquet

Cloud & DevOps: AWS Lambda, S3, SageMaker, ECR / ECS, AWS CDK, Docker, IAM, VPC / Network Security, CI/CD

EXPERIENCE

Data Scientist (Geospatial & AI)

Apr 2025 – Aug 2026

Florida Power & Light - Contract, West Palm Beach, FL

- Built and deployed an overhead switch-state (open/closed) classification model, training it locally then migrating to AWS (ECR, SageMaker) for production inference.
- Used Roboflow for annotation and data preparation on both production CV models, including a 6x augmentation pipeline (flips, brightness, and other transforms), train/test/validation splits, and YOLO-format export.
- Contributed to testing and evaluating CV architectures (YOLO v8/v9/11, Faster/Mask R-CNN, EfficientDet, SSD, DETR/Deformable DETR, Detectron2, MMDetection, RT-DETR) against a precision/recall/F1 and image-scale evaluation methodology co-designed with the team, producing statistical evaluation reports for leadership; selected RT-DETR for transfer learning and standardized the approach for vendors.
- Co-developed an equipment detection and segmentation model (with a data scientist and software engineer) spanning 27-30 asset classes, evolving from object detection to segmentation.
- Evaluated ~20 vendor bids with a GPT-based scoring agent, then audited 3 external ML vendors post-award for licensing compliance and supported resolving CUDA/GPU and batch-processing mismatches against AWS SageMaker.

Associate Geospatial Analyst

Sep 2024 – Mar 2025

Florida Power & Light - Contract, Jupiter, FL

- Led QA/QC of LiDAR-derived power grid data for infrastructure resilience work, including pole hardening.
- Managed PostgreSQL/PostGIS databases, optimized Cloud-Optimized GeoTIFFs, and automated QA workflows in FME, ArcPy, and Model Builder to streamline data storage and validation.

Research Assistant

Jan 2023 – Aug 2024

Florida Atlantic University, Boca Raton, FL

- Conducted FEMA hydrologic and hydraulic (H&H) flood risk assessments for 3 Florida communities using Python automation, part of FAU's \$1.7M FEMA-funded CWR3 Watershed Master Planning Initiative to build a statewide planning template.
- Built a Python geocoding tool processing 3 million addresses, replacing a paid service; cut turnaround from 3 months to 1 week and saved \$20K in licensing.

Geomatics Engineer

Dec 2020 – Dec 2022

Freelance Consultant, Kathmandu, Nepal

- Led survey teams for municipal road, control, and construction surveys, including GNSS landmark and UAV topographic/transmission-line surveys, and drone photogrammetry for orthomosaic mapping.

EDUCATION

MSc, Geosciences — GIS Concentration — Florida Atlantic University · 2023–2024

BSc, Geomatics Engineering — Tribhuvan University, Nepal · 2016–2021

CERTIFICATIONS & TRAINING

- The Rise of the AI Data Engineer Bootcamp (2026)
- FAA Part 107 — Remote Pilot Certificate
- Machine Learning Specialization — DeepLearning.AI
- Deep Learning Specialization — DeepLearning.AI
- Remote Sensing Image Acquisition & Analysis — UNSW Canberra
- **NASA ARSET Certifications (12):** Applied remote sensing training (2022-2024) in climate, disaster, and environmental risk modeling.
- **Esri Training Certifications (8):** GIS fundamentals, cartography, spatial data science, and imagery analysis.
- GIS Mapping & Spatial Analysis — U. Toronto
- Data Science with R — Johns Hopkins
- GIS Specialization — UC Davis
- Introduction to Databases — Meta

PUBLICATIONS

Dhungana, B.; Liu, W. *Urban–Rural Exposure to Flood Hazard and Social Vulnerability in the Conterminous United States*. ISPRS Int. J. Geo-Inf. 2024, 13, 339.

Mandal, A.K.; Thapa, A.K.; Thapa Chhetri, M.; Dhungana, B.; Bloetscher, F.; Yong, Y.; Su, H. *PyWMP: A Unified Python Framework for Multi-Fidelity 1D, 2D, and Hybrid Event-Based Flood Modeling*. Manuscript in review, DOI pending; ongoing contributor to the open-source PyWMP package.

PROJECTS

wildfire-copilot: A LangGraph agent for utility wildfire risk — combines a trained XGBoost susceptibility model with live NASA/NWS hazard feeds and RAG over real wildfire-mitigation filings to answer plain-English risk questions and prioritize inspections. (*LangGraph, XGBoost, RAG, FastAPI*) [Demo](#) | [Repo](#)

geomlint: A CI-first data-quality linter for vector geospatial files — catches invalid geometry, CRS confusion, and topology errors that mainstream data-observability tools skip. No GDAL required; pip install and it runs anywhere pip does. (*Python, Shapely, PyProj, PyPI, CLI*) [Docs](#) | [Repo](#)

Geospatial Data Copilot: Ask a plain-English question about San Francisco's utility infrastructure — a local LLM agent (LangGraph ReAct, Ollama) translates it to spatial SQL, runs it, and answers with a summary and interactive map. No API keys. (*LangGraph, Ollama, DuckDB Spatial, Chroma*) [Demo](#) | [Repo](#)

HazardMCP: An MCP server unifying FEMA, USGS, NOAA, NWS, and NASA hazard data behind a single AI-agent interface — one-call location risk summaries and map rendering, compatible with Claude Desktop. (*Python, FastMCP, Folium, Docker*) [Repo](#)

Conversational AI Assistant: LangChain RAG chatbot with a history-aware retriever for multi-turn, context-grounded conversation, citing sources. Runs locally via Ollama. [Repo](#)

RomanNepaliAI: Converts romanized Nepali to Devanagari live, client-side, plus a CLI that batch-translates SRT subtitles through pluggable backends. [Demo](#) | [Repo](#)